HW4: Probabilistic Clustering and Gaussian Mixture Models

MA 5755 Data Analysis & Visualization in R/Python/SQL
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Introduction

This assignment revisits the Iris dataset with a focused objective: to understand how distance-based and model-based clustering methods differ in the structure they impose on data, and in the information they yield about that structure. To make these differences as visible as possible, the analysis is restricted to the two overlapping species, *Iris versicolor* and *Iris virginica*, and to two features, petal length and petal width, both standardised to zero mean and unit variance.

Figure 1 establishes the ground truth. Versicolor (blue) occupies the lower-left region of the feature space, with relatively shorter and narrower petals, while Virginica (red) occupies the upper-right, with longer and wider petals. Critically, the two clouds overlap substantially in the diagonal band.

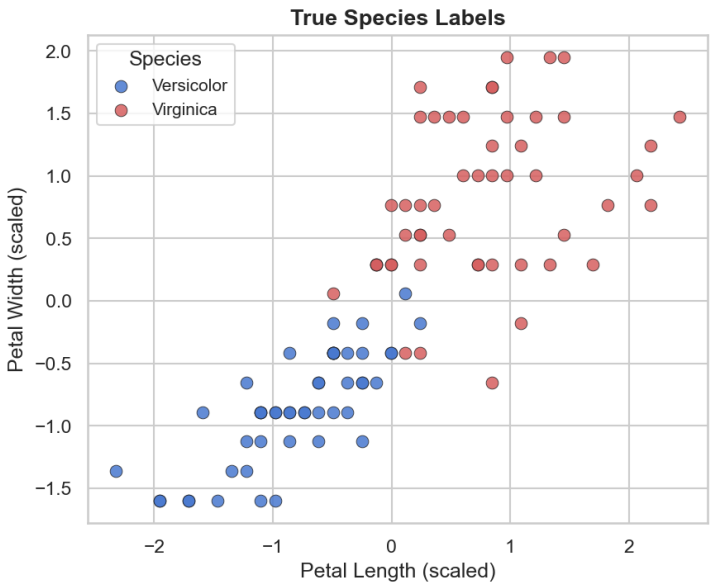

Figure 1: Scatter plot of the two standardised petal features for *Iris versicolor* (blue) and *Iris virginica* (red). The substantial overlap in the central diagonal band and the positive correlation between petal length and width are the two structural challenges that each clustering method must handle.

Task 1: k-Means vs Spherical GMM vs Full-Covariance GMM

Figure 2 places the three fitted models side by side, each panel showing cluster assignments and the red-star representative (centroid for k-Means, component mean for each GMM variant).

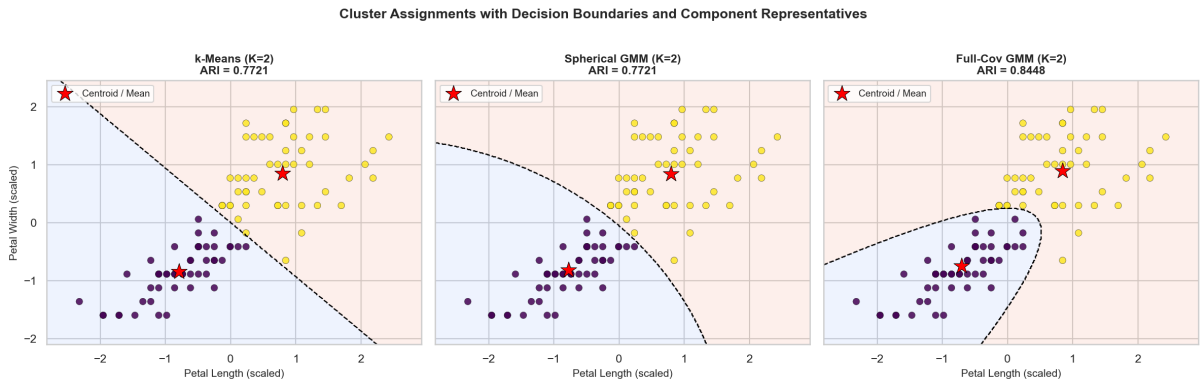

Figure 2: Cluster assignments and decision boundaries for k-Means (left), Spherical GMM (centre), and Full-Covariance GMM (right), each with $K = 2$ on the standardised petal features. Red stars mark the centroids or component means. Shaded regions show the partition of the feature space; dashed lines show the decision boundary. The ARI against true species labels is shown in each panel title.

The two red stars in the left and centre panels are nearly co-located, confirming that

the centroids and spherical GMM means converge to approximately the same positions. Both methods misclassify (comparing with the Figure 1) a portion of the points in the overlapping diagonal zone, because neither has a mechanism to represent the elongated, correlated shape of the two species clouds.

k-Means and Spherical GMM: Similar but Not Identical

k-Means partitions data by assigning each point to its nearest centroid under Euclidean distance, which is mathematically equivalent to assuming spherical Gaussian components with equal variance $\sigma^2\mathbf{I}$. The resulting boundary is a straight hyperplane, the perpendicular bisector of the line joining the two centroids, visible as the diagonal dashed line in the left panel of Figure 2.

The spherical GMM imposes the same $\sigma_k^2\mathbf{I}$ constraint, but crucially, each component is allowed to fit its own σ_k^2 independently. Because the two fitted variances are not exactly equal, the decision boundary curves slightly relative to k-Means, as visible in the centre panel. But this curvature is insufficient to change any hard label assignments, and both methods produce an ARI of **0.7721**. The two red stars in the left and centre panels are nearly co-located, confirming that the centroids and spherical GMM means converge to approximately the same positions. This near-equivalence illustrates that under approximately spherical assumptions, adding probabilistic mechanism without relaxing the covariance structure yields no practical improvement in label recovery.

Full-Covariance GMM: An Elliptical Boundary

The right panel reveals a qualitatively different geometry. With an unconstrained covariance matrix Σ_k, the full GMM no longer produces a single dividing curve , instead, the decision boundary forms a closed elliptical loop enclosing the lower-left (Versicolor) cluster. This occurs because the full GMM fits an elliptical Gaussian to each component, and the boundary between two such ellipses is in general a closed conic section rather than a half-plane. The enclosed contour traces the region where the Versicolor component's density exceeds the Virginica component's density, adapting to the elongated, correlated shape of the data.

The component mean in the right panel is also positioned differently from the k-Means centroid, reflecting that EM maximises log-likelihood rather than minimising sum-of-squared distances , two objectives that diverge when the covariance is unconstrained. This geometric flexibility yields an ARI of **0.8448**, a meaningful improvement over both spherical methods.

Task 2: Soft vs Hard Assignments

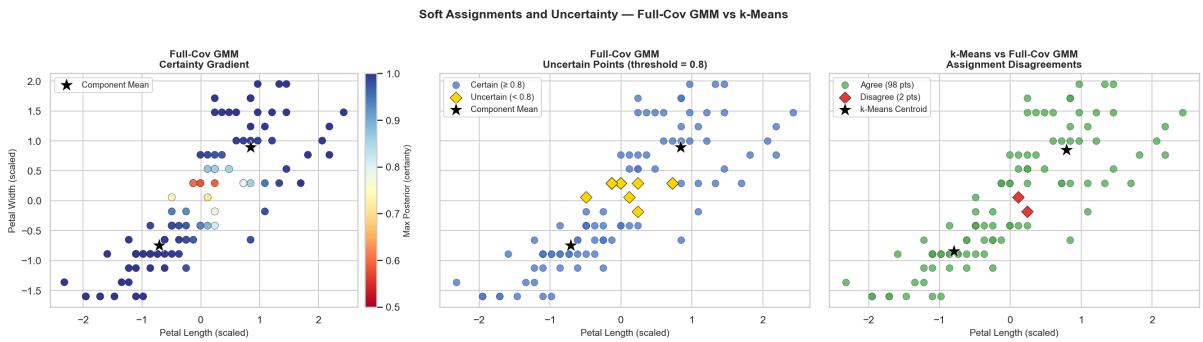

Figure 3: Three-panel uncertainty analysis using the full-covariance GMM. *Left:* Continuous certainty gradient where each point is coloured by $\max_k \gamma_{ik}$, from red (≈ 0.5, uncertain) to blue (≈ 1.0, certain). *Centre:* Points with max posterior < 0.8 flagged as uncertain yellow diamonds (5 points); the remainder shown in blue. *Right:* Agreement between k-Means and full GMM assignments: green where both methods agree (98 points), red diamonds where they disagree (2 points). All disagreements fall within the uncertain band identified in the centre panel.

Figure 3 investigates soft assignments and uncertainty through three complementary views.

The first panel maps the full GMM's posterior certainty $\max_k \gamma_{ik}$ as a continuous colour gradient from red (≈ 0.5, maximally uncertain) to blue (≈ 1.0, fully certain). Uncertainty is tightly localised, the red and orange points form a narrow diagonal strip in the central overlap zone, while the majority of observations are deep blue, indicating high-confidence assignments well away from the boundary.

The second panel discretises this with a threshold of 0.8, flagging only **5 observations** as uncertain. These yellow diamond markers are spatially confined to a small, neighbourhood of the decision boundary, confirming that the GMM's ambiguity is both meaningful and geographically specific.

The third panel: each point is coloured green if k-Means and the full GMM assign it to the same cluster, and red if they disagree. Despite the qualitatively different decision boundaries, a straight Voronoi line for k-Means versus a closed elliptical contour for the full GMM (Figure 2), the two methods disagree on only **2 out of 100 points**, both of which lie within the uncertain band identified in the second panel. k-Means arrives at the same assignment as the GMM for 98 points, but does so without any mechanism to signal that 5 of those points are ambiguous.

Conclusion

Taken together, the three tasks illustrate a progression from geometry to probability. k-Means' implicit assumption of spherical, equal-variance clusters produces a Voronoi partition that cannot represent the correlated, elongated structure of the petal feature space, and is incapable of expressing any uncertainty about its assignments.

Adding probabilistic machinery without relaxing the covariance constraint, as in the spherical GMM, yields no practical improvement: the ARI remains identical at 0.7721, and the boundary curves only marginally. Only when the covariance is freed does the model gain the geometric flexibility to produce a qualitatively different boundary, a closed elliptical contour rather than a hyperplane, and to better recover the true species structure (ARI = 0.8448).

Despite their visibly different boundaries, k-Means and the full GMM disagree on only 2 out of 100 assignments. The GMM identifies 5 genuinely ambiguous observations and expresses graded uncertainty across the overlap zone; k-Means reaches nearly the same decisions but does so with no mechanism to signal where the evidence is weak.

In this sense, k-Means is best understood as a special case of a GMM: one with spherical covariances, equal mixing weights, and hard-assignment approximations to the EM E-step. For data with correlated features and overlapping densities, as is the case here with *Iris versicolor* and *Iris virginica*, the richer probabilistic framework of the GMM is a better strategy for looking at these data.